H^0(U, R SHEAFHOM(L,M)) = HOM_D(L|_U, M|_U)

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Allowed standard prerequisites (editorial).

- Hom complex into K-injectives is K-injective. Cohomology of Sheaves, Lemma cohomology-lemma-RHom-from-K-flat-into-K-injective. SheafHom^•(L^•,I^•) is K-injective.
- Sections of the hom complex compute derived Homs. Cohomology of Sheaves, Lemma cohomology-lemma-RHom-into-K-injective. H^0(Γ(U, SheafHom^•)) = Hom_D(L|_U, M|_U).

Source reference. Original lemma 0A8T: sources/stacks-chow/tags.txt, source lines 12926–12926. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0A8R: sources/stacks-chow/tags.txt, source lines 12924–12924. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let L, M be objects of $D(\mathcal{O}_X)$. For every open U we have*

$$H^0(U, R\mathcal{H}om(L, M)) = \mathrm{Hom}_{D(\mathcal{O}_U)}(L|_U, M|_U)$$

and in particular $H^0(X, R\mathcal{H}om(L, M)) = \mathrm{Hom}_{D(\mathcal{O}_X)}(L, M)$.

Proof. Choose a K-injective complex $\mathcal{I}^\bullet$ of $\mathcal{O}_X$-modules representing M and a K-flat complex $\mathcal{L}^\bullet$ representing L. Then $\mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{I}^\bullet)$ is K-injective by Lemma 0A8T. Hence we can compute cohomology over U by simply taking sections over U and the result follows from Lemma 0A8R. □

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.